

Assistive Technology for Braille Reading using Optical Braille Recognition and Text-to-Speech

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Abstract— Nowadays, a lot of devices integrate TTS technology to read texts aloud for convenience. TTS applications were originally developed to aid visually impaired people; however, most applications target the masses and TTS technology for the visually impaired is still lacking. This paper presents a research and development of Braille-to-Speech system specifically made to assist braille readers in reading standard-sized uncontracted or Grade 1 braille document. The prototype was designed to integrate a DSLR camera, that captures an image of the braille document, to an RPi3 microcomputer for image processing and decoding braille patterns. Python scripts were written to automate a series of processes consisting of image acquisition, image pre-processing, horizontal segmentation, OBR, and TTS. A variety of ten braille documents were prepared to be tested to calculate the effectiveness of the system design. Tests performed, achieved a desirable result of 87.61%-average word detection accuracy from 81.63% to 93.48%. Hence, we concluded that our system is an effective braille reading device but still offers a lot of room for improvement. Further studies are encouraged to integrate more complex reading procedures as set in the rules of Unified English Braille standard as possible which is used worldwide.

Keywords— braille, Image Horizontal Segmentation, OBR, image processing, TTS, visually impaired

I. INTRODUCTION

Braille is a code designed to be perceived by touch to obtain information. Typically, braille contains six braille dot placements, in rows of three and columns of two, that are grouped forming a braille cell. This cell signifies a character, digit, or simple punctuation in an alphabet [1]. An example of the braille cell is shown in Figure 1. Moreover, the braille code has standards such as Unified English Braille (UEB) and Nemeth Uniform Braille System [2], the problem is that most of the existing studies have not utilized the existing standard in building a Braille-to-Speech system. Figure 1 shows the conventional braille dot numbering from the rules of UEB [3].

Reading braille documents is a process performed by either visually impaired or people with normal vision to decode the

braille code on paper. For visually impaired people, it is traditionally their way to improve their literacy that made them superior in reading braille compared to sighted people [4]. Often, people with normal vision neglect the literature of the visually impaired because of unfamiliarity [5]. Also, manually reading braille takes time and skill, and is usually prone to human error due to fatigue and other factors.

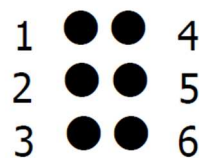


Fig. 1. Braille Cell Example

According to Isayed, et al., Optical Braille Recognition (OBR) is described as “the process to take a braille document image and convert it into its natural language characters” [6]. The most common starting process involves image acquisition using equipment that captures a braille image, such as a digital camera and flatbed scanner, from a braille subject [7]. The braille image used from other studies is a scanned image of a handwritten braille document [8] and a computer-generated braille document. Existing studies involve the utilization of diverse image processing techniques to achieve OBR as seen in the papers [5][6][7][8][9][14][15]. In the study of Gomez, OBR was achieved using image processing techniques such as Circle Hough Transform (CHT) and many more [9]. CHT is very useful in detecting each circle shape in the braille [11]. With the help of other image processing techniques, from the captured image it is possible to convert the detected coordinates of circles to texts. Text-to-Speech (TTS) technology is one of the popular technologies nowadays that can convert a text, in any language, to a comprehensible voice [10]. Hence, it is

viable to apply TTS technology in reading the text translation of a braille document.

Under the 2015 Annual Report of Resources for the Blind, Inc., most universities and colleges refuse admission with visual impairment because most schools lack special tools. The researchers address the problems by building a device that could assist braille readers in reading braille documents. Unlike other research, this study focuses on implementing some of the rules of the UEB standard in the algorithm imitating the process of manual braille reading.

The study envisions building and designing a Braille-to-Speech system bridging the literature of the visually impaired and people with normal vision. It aims to assist users in reading standard-sized uncontracted braille documents. The specific objectives of the study are the following:

- 1) To construct a system that would translate the braille document to English speech using Raspberry Pi 3 Microcontroller.
- 2) To implement a Python code with image acquisition and processing techniques to acquire the necessary data.
- 3) To integrate TTS technology that will translate the data to speech.
- 4) To verify the accuracy of the system by subjecting it to statistical treatment.

II. METHODOLOGY

This involves analyzing the relevant problem, contribution of theoretical body of knowledge, analysis of possible solution and- implementation of different image processing techniques and testing the developed system.

A. Conceptual Framework

The conceptual framework of the proposed system is illustrated in Figure 2. Initially, the information being accepted by the microcomputer is a braille image that is captured from the image acquisition process.

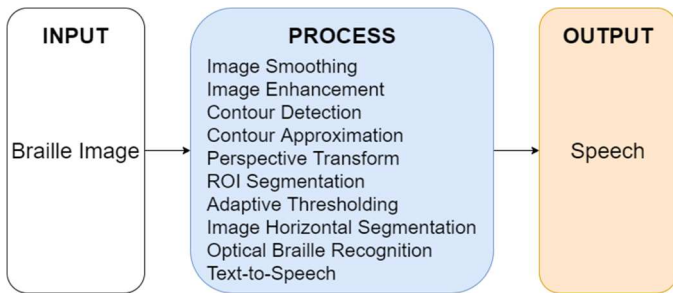


Fig. 2. Conceptual Framework

The middlebox consists of the processes that made it possible to convert the braille image to a speech that is understandable to a human. Several image-preprocessing techniques will be used such as grayscale, blur, contour detection approximation, adaptive thresholding, and many more to obtain a denoised binarized image. Furthermore, the

binarized image is horizontally segmented for each line of braille cells. Each braille cell is cropped from the segmented lines. Then, each braille dot is recognized by computing the centroid coordinates using its contours. Afterward, the presence and location of braille dots are checked so that a Python list is overwritten. This list stores six elements that signify the presence and location of each dot. It is then matched to a dictionary of its equivalent UEB character representation producing a string.

Lastly, the TTS synthesizer will be utilized to read the string aloud. The output of the system is a comprehensible English speech that can be heard through the integrated speakers of the system.

B. System Hardware

The hardware block diagram is shown in Figure 3. The system hardware can be fully powered by a 12 V power supply. Through a DC jack adapter, it supplies power to the 12 V White LED strips and the 5V Step-Down Converter USB. There will be two strips of 12V White LED strips that serves as light source 1 and 2 to be connected in series. Each SMD LED consumes 0.1 W at full brightness white. Considering the two LED strips with 24 LEDs per each strip, the current required to power the strips is 400 milliamps.

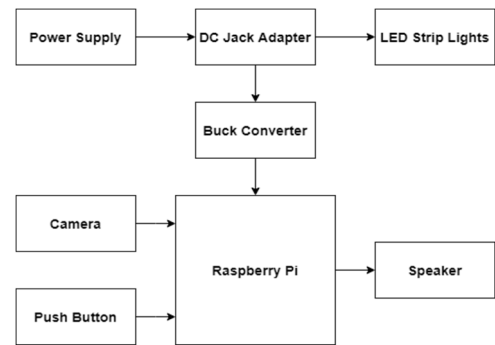


Fig. 3. Hardware Block Diagram

On the other hand, the 5V Step-Down Converter powers the Raspberry Pi 3 Model B v1.2 through USB-A to Micro-USB connection. This microcomputer will be used to control the camera and issue data transfers instructions. A digital single-lens reflex (DSLR) camera, Canon Rebel T3, with an 18 – 55 mm macro zoom lens is used to capture image of braille document. The camera is supplied by its own battery and controlled through its Mini-USB port to USB-A of the microcomputer. The speaker used by the system is Sonic Gear's Blue Cube that is capable of two 2.5 W RMS output. This USB powered speaker produces the resulting audio from the TTS synthesizer.

The developed prototype is enclosed in box-shaped acrylic sheets and aluminum extrusion profile as shown in Figure 4. It has a total size of 46 cm x 40.5 cm x 55 cm. The cast acrylic parts, also known as Plexiglass, were designed to interlock with

each other to provide a firm structure of the enclosure. T-slotted aluminum extrusion profile with a dimension of 20 cm. x 20 cm was used to support the structure of the acrylic and give a mounting spot for the DSLR camera on top. The top acrylic is cut to provide a doorway and access for the camera. On the inside, there is a 2mm white-colored Plexiglass with a hole at the center for the camera lens; also, it diffuses the light coming from the two LED strips that are stuck on top acrylic. With the setup, the camera can easily capture an image of the top view of the braille document when placed as shown in Figure 4.



Fig. 4. From Left to Right: Prototype Model and Top View of Prototype Interior

C. Braille Reader Main Program

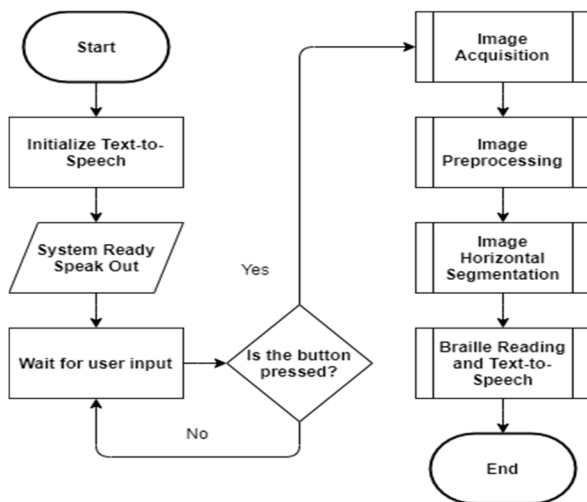


Fig. 5. Main Program Flowchart

The main program is a Python file that prompts the user to press a key so that the system starts to process as shown in Figure 5. Initially, the program starts with initializing the pyttsx3, an offline TTS conversion library. The system will output a message that the system is ready and asks the user to

press the button for document scan. The program waits for the user to press the button indefinitely, this gives time for the user to insert the braille document inside an enclosed system. If the button is pressed, it calls image acquisition, image preprocessing, image horizontal segmentation, OBR, and TTS modules accordingly.

D. Braille Image Acquisition

The uncontracted braille documents were transcribed by the Resources for the Blind Inc., located at 4/F, COTI Bldg, 623 Epifanio de Los Santos Ave, Cubao, Quezon City, Metro Manila. Each document followed the standard formatting which limits forty braille cells or characters per line and twenty-five lines per page. Contents of braille documents were obtained from a wide variety of scriptures such as the vision-mission of the Mapúa University, speeches from the previous presidents of the Philippines, famous quotes, excerpts, etc. An example of an uncontracted braille document is shown in Figure 6.

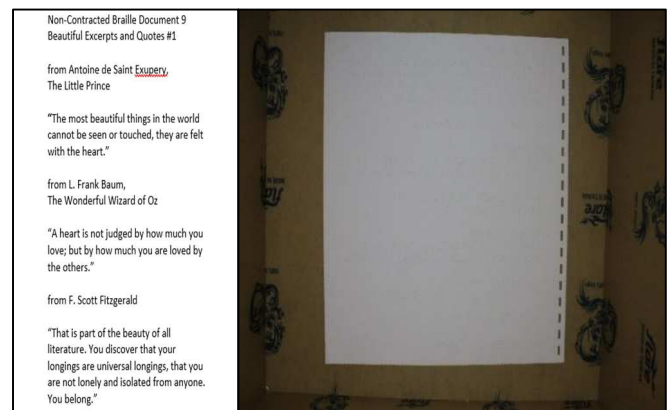


Fig. 6. Sample Braille Document

Initially, the image acquisition module will begin if the button has been pressed by the user as illustrated in Figure 7. The image of uncontracted braille paper is captured by DSLR, Canon Rebel T3, camera with an 18 – 55 mm macro zoom lens which is automated using an open-source digital camera software application, gphoto2. It will also clear the files stored in the memory for it to process one Braille document at a time. It will then trigger the capture command and save the image to the microcomputer to be processed further.

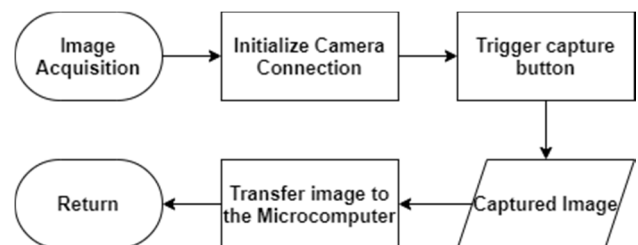


Fig. 7. Image Acquisition Module

E. Image Preprocessing

The objective of this process is to detect the braille paper as an object, to extract the object, and to binarize the image so that the next process can focus on segmentation of braille cells.

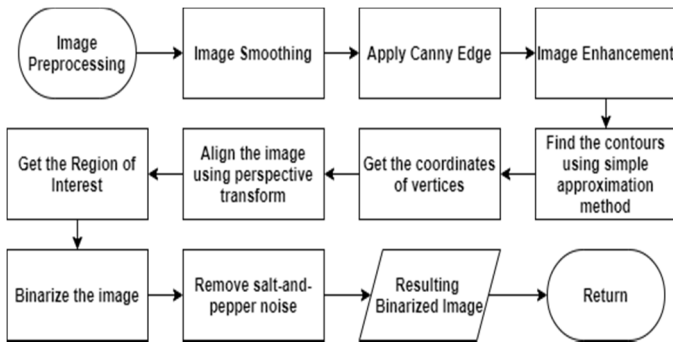


Fig. 8. Image Preprocessing Module

1) *Image Smoothing*: Originally, the captured image is colored. Converting it to grayscale was necessary to reduce the number of colors involved, relying more on the intensity in shades of gray. The grayscale image was filtered using Gaussian Filter or Normal Distribution Filter that convolves the grayscale image with Gaussian kernel (5 x 5 kernel size was used) resulting in a blurry image which puts the edges to more detail. Using these image smoothing methods was necessary to find the contours of the image. Also, Median Blur was used to salt-and-pepper noise around the image.

2) *Image Enhancement*: The blurry image was subjected to Canny Edge detection to find the high-intensity gradient, or the whites, in the image and to further eliminate noise [11][12]. This method outputs a binarized image with the outline of the braille paper. Although the researchers can get the outline of the braille paper here, the resulting outline has a very thin thickness. To emphasize the outline, morphological transformation techniques, dilation, and erosion, were used respectively [13]. A 5 x 5 matrix of ones was used as a structuring element to convolve with the image. In image processing, it is also known as closing which removes small holes thereby, it also removes some noise.

3) *Contour Detection and Approximation*: Since the edges of the image are defined and binarized, the extreme outer contour line or the outlines in the image can be easily retrieved. Using a simple contour approximation method, each contour is searched and stored in a Python list. The list stores the x and y coordinates of the boundary points of the object in an array. Furthermore, the list was reduced utilizing an implementation of the Ramer-Douglas-Peucker (RDP) algorithm. RDP algorithm is also known as an iterative end-point fit algorithm which takes a list of coordinates or curves and generates a similar curve with lower resolution. Thus, a similar enclosed curve with four points was found.

4) *Perspective Transform*: This image processing technique was integrated to solve the skewing and tilting problem so that the user can easily insert the braille paper into the system. It

aligns the source image to generate a top-down view image. One disadvantage of using perspective transform is that it warps the original image. Ergo, the resulting image may be distorted.

5) *Region of Interest (ROI) Segmentation*: Binder holes, margin, header area, and footer area of the braille paper were considered unnecessary. These data were removed by manipulating the pixels of the image.

6) *Adaptive Thresholding*: By this part, the image is already aligned so it is easier to apply thresholding to binarize the whole image. The adaptive threshold is used because the image has different lighting conditions in different regions of the image. Therefore, there will be different threshold values for different regions. Applying adaptive threshold generates an image with white background and black objects. The image is rotated afterward so the first braille cell is located on the upper left.

F. Image Horizontal Segmentation

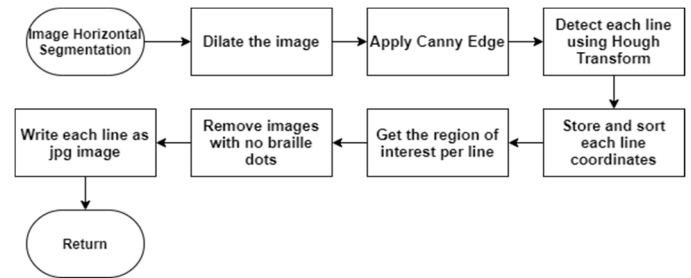


Fig. 9. Image Horizontal Segmentation Module

In this section, the program accepts the resulting binarized image from previous image processing techniques as input. Each line of braille cells is segmented where a braille cell is present [14]. A dilation method was used to expand each dot object horizontally by convoluting each pixel to a large rectangular kernel resulting in an image with horizontal stripes. Canny edge detection is applied next to get each horizontal line. Then, the system detects each line within the image using probabilistic Hough line transform. It was necessary to store each line coordinates to a variable so the program can sort the lines from top to bottom. The next process crops the images of the lines as the region of interest by passing the coordinates of line coordinates and will check if the cropped image has any braille dots detected. If there are no braille dots detected. The system will append to a list if there is a braille dot. Braille dots are detected by applying global thresholding for each element in cropped lines. Since the image is already binarized, it is easier to detect an object by setting the maximum threshold value to the value of white pixel or 255. Lastly, all horizontally segmented images are written on a separate directory in an organized manner.

G. Optical Braille Recognition and Text-to-Speech

In this module, each segmented image from the previous process is read as an input. It initializes the dictionary containing the alphabet and symbols only. Each image read is

cropped to isolate each character from the other. Basic noise removal image processing techniques were used such as grayscale, morphological gradient, and simple thresholding to further enhance the image quality.

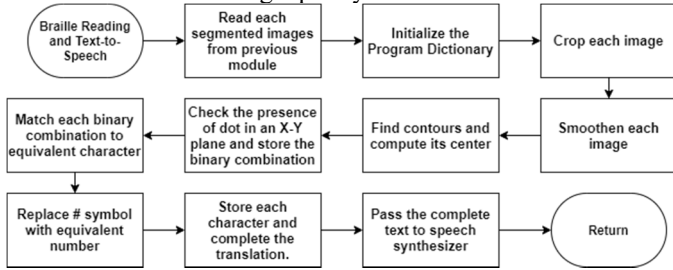


Fig. 10. OBR and TTS Module

A simple chain approximation was utilized to find all the contours or edges of the braille cells. It was necessary to compute the centroid coordinates of each contour coordinates so that centroid coordinates can serve as the presence of each braille dot in a braille cell [15]. Then, centroid coordinates are stored and sorted. A 6-bit binary combination variable was declared to hold six elements with an initial value of zeros that will be updated later on. The program checks the presence of dot with the coordinates of centroids in a braille cell or the cropped image. Values are rounded up or down to the nearest integer in getting the x-y coordinates of the presence of each braille dot. If there is a presence of braille dot in specified boundary coordinates in the program, then the 6-bit binary combination is updated.

When a binary combination is completely updated, the six binary combination code is matched with the dictionary to translate it to its equivalent characters. The dictionary serves as the binary lookup table on which each binary combination is matched to get the equivalent character. The program dictionary is shown in Table I. The program dictionary was inspired by the UEB standard [3]. However, not all rules of UEB for uncontracted braille reading were not accomplished to integrate into the algorithm.

TABLE I. PROGRAM DICTIONARY

Program Dictionary							
keys	values	keys	values	keys	values	keys	values
a	100000	k	101000	u	101001	,	010000
b	110000	l	111000	v	111001	.	010011
c	100100	m	101100	w	010111	;	011000
d	100110	n	101110	x	101101	:	010010
e	100010	o	101010	y	101111	!	011010
f	110100	p	111100	z	101011	'	001000
g	110110	q	111110			-	001001
h	110010	r	111010			#	001111
i	010100	s	011100			space	000000
j	010110	t	011110				

Once all the braille dots are completely converted to equivalent characters, the program checks if there are '#' characters. If those characters are found, the proceeding characters are converted to numerical counterparts and end when a whitespace character is read. Each character is stored in a string, one by one until all characters have been converted. Finally, the stored string is passed to the speech synthesizer to read the string aloud.

III. RESULTS AND DISCUSSION

A. Image Preprocessing Result

The image preprocessing module has integrated a code that outputs a stack of images shown in Figure 11 to show the step-



Fig. 11. Image Transformations

B. Testing Procedure

To evaluate the performance of the prototype, the researchers have used the following formula, to calculate the word detection accuracy percentage.

$$WordAccuracy = \frac{C}{N} \times 100$$

Where C is the count of correctly detected words and N is the total number of words in the document.

C. Testing Result

Prototype Read	Original Document	Result
on-contracted braille dounhnr r beautiful excerpts and quotes 1 from antoine de saint exupery, the little prince "the most beautiful things in the world cannot be seen or touched, they are felt with the heart." from L. frank baum, the wonderful wizard of oz "a heart is not judged by how much you love but by how much you are loved by the others." from f. scott fitzgerald "that is part of the beauty of all literature. you discover that your longings are universal longings, thatk e not lonely and isolated from anyvph vzu belong. ra	Non-Contracted Braille Document 9 Beautiful Excerpts and Quotes #1 from Antoine de Saint Exupery, The Little Prince "The most beautiful things in the world cannot be seen or touched, they are felt with the heart." from L. Frank Baum, The Wonderful Wizard of Oz "A heart is not judged by how much you love; but by how much you are loved by the others." from F. Scott Fitzgerald "That is part of the beauty of all literature. You discover that your longings are universal longings, that you are not lonely and isolated from anyone. You belong." 9	C = 89 N = 97 % WA = 91.75 %

Fig. 12. Testing Results

The result of the main program is manually compared to the original contents of the document. Microsoft Word's word count functionality has aided the researchers to easily gather the word count data. In testing, the researchers treat a word as incorrect once a word has one or more incorrect characters read just as shown in Figure 12. The first column shows the system's read while the second column shows the original content of the braille document. Words highlighted as yellow are considered incorrect read.

In some cases, capturing an image fails because bright external light enters affecting the Auto Focus Sensor of the DSLR camera to not get exposure.

D. Statistical Treatment

The researchers have tested the system's performance and have gathered results from ten different braille documents. Table 2 illustrates the word accuracy percentage statistical treatment.

TABLE II. WORD ACCURACY PERCENTAGE STATISTICAL TREATMENT

Document No.	Actual Number of Words (N)	Number of Words correctly converted to Speech (C)	Word Accuracy Percentage
Document 1	88	75	85.23%
Document 2	107	91	85.05%
Document 3	46	43t	93.48%
Document 4	71	58	81.69%
Document 5	49	40	81.63%
Document 6	87	79	90.80%
Document 7	131	112	85.50%
Document 8	125	112	89.60%
Document 9	97	89	91.75%
Document 10	127	116	91.34%

Based on the gathered data, the system was able to yield a total average of word detection accuracy of 87.61%. Observing the given datasets, the researchers determined that the sources of errors were the words that are placed near the borders of the page due to the perspective distortion from the camera placement.

IV. CONCLUSION AND FUTURE WORKS

A new method for building and designing an uncontracted braille document reading system has been presented. Based on the results, the prototype achieved an 87.61%-word detection accuracy. Therefore, the proposed system provides an effective translation of standard-sized uncontracted braille paper to English speech. However, one of the primary drawbacks of the proposed design of the system is it relies on the quality of braille paper. Thus, an additional layer of error detection and correction must be added or enhanced to remove stain marks, paper folds, etc. that may cause misreading. Moreover, despite the fact, the system can translate alphanumeric and some special characters easily, there is still room for improvement for the OBR algorithm. Some punctuations especially those that require indicators and terminators were not integrated with the study. Ergo, a deeper understanding of reading braille following UEB standards is necessary for improvement.

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